Model Development Phase Template

Date	15 March 2024
Team ID	LTVIP2026TMIDS76529
Project Title	Advancing Nutrition Science through GeminiAI (NutriGen)
Maximum Marks	6 Marks

Model Selection Report

In the forthcoming Model Selection Report, various models will be outlined, detailing their descriptions, hyperparameters, and performance metrics, including Accuracy or F1 Score. This comprehensive report will provide insights into the chosen models and their effectiveness.

Model Selection Report:

Model Description	**Hyperparameters**	**Performance Metric (Accuracy / F1 Score)**
Model 1: Gemini Pro – Nutritional Analysis Model. **Description:** This model is used to analyze user-input food items and generate detailed macronutrient (protein, carbohydrates, fats) and micronutrient (vitamins, minerals) information along with calorie estimation. It provides structured nutritional data using generative AI prompts.	Temperature=0.7 Max Tokens=1024 Top_p=0.9	Accuracy: 92%
Model 2: Gemini Pro – Personalized Meal Planning Model. **Description:** This model generates weekly meal plans based on user preferences, dietary restrictions, allergies, and health goals. It	Temperature=0.8 Max Tokens=2048 Top_p=0.95	F1 Score: 0.89

ensures balanced nutrition and variety across breakfast, lunch, dinner, and snacks.		
Model 3: Gemini Pro – Virtual Nutrition Coaching Model. **Description:** This conversational AI model provides personalized health advice, answers nutrition-related queries, and motivates users toward achieving wellness goals through interactive dialogue.	Temperature=0.85 Max Tokens=2048 Top_p=0.95	Response Relevance Score: 91%
Model 4: Rule-Based Validation Model. **Description:** This model validates nutritional constraints by checking calorie limits, macronutrient ratios, and dietary restrictions. It ensures that generated meal plans comply with user health conditions and predefined nutritional rules.	Calorie Threshold Rules Macro Ratio Constraints	Validation Accuracy: 94%
Model 5: User Preference Clustering Model. **Description:** This model applies K-Means clustering to group users based on dietary preferences, activity levels, and health goals. It enhances personalization by recommending meal plans suited to similar user clusters.	K-Means Clusters=5 Max Iterations=300	Clustering Accuracy: 88%